

QudoSSL Commercial — Performance (macOS arm64)

Unmodified upstream **OpenSSL 3.5.7** — the QudoSSL Commercial product (no Community/delegated build involved). Measured with OpenSSL’s own `speed` and `perftools` handshake tools, median of 6 repetitions, single core. Higher is better.

Environment

CPU	Apple M4
Architecture	arm64
SIMD	neon
Cores	10
OS	macos
Kernel	Darwin 25.5.0
OpenSSL	OpenSSL 3.5.7 9 Jun 2026 (Library: OpenSSL 3.5.7 9 Jun 2026)

Primitives — FIPS provider (`openssl speed`, operations / sec)

Key encapsulation — ML-KEM and hybrid groups

Group	keygen	encapsulate	decapsulate
ML-KEM-512	37,832	162,415	104,838
ML-KEM-768	25,390	110,618	72,993
ML-KEM-1024	17,513	78,903	53,515
X25519MLKEM768	18,885	25,926	33,326
SecP256r1MLKEM768	10,028	10,202	22,714
SecP384r1MLKEM1024	554	478	2,636

Signatures — ML-DSA

Algorithm	keygen	sign	verify
ML-DSA-44	2,989	4,160	21,433
ML-DSA-65	1,795	2,543	13,912
ML-DSA-87	1,473	2,172	8,783

Primitives — default provider (`openssl` speed , operations / sec)

Key encapsulation — ML-KEM and hybrid groups

Group	keygen	encapsulate	decapsulate
ML-KEM-512	96,771	160,008	104,620
ML-KEM-768	61,303	110,372	72,286
ML-KEM-1024	40,462	78,226	53,230
X25519MLKEM768	32,162	25,590	32,741
SecP256r1MLKEM768	39,080	21,241	22,330
SecP384r1MLKEM1024	2,516	1,347	2,582

Signatures — ML-DSA

Algorithm	keygen	sign	verify
ML-DSA-44	23,262	4,162	21,468
ML-DSA-65	12,263	2,563	13,980
ML-DSA-87	9,131	2,157	8,835

TLS 1.3 handshake (`perftools` , in memory, single core, ML-DSA-44 certificate)

Provider	Key exchange	μ s / handshake	Handshakes / sec
fips	X25519MLKEM768	586.4	1,705
default	X25519MLKEM768	556.3	1,797
default	SecP256r1MLKEM768	582.1	1,718
default	X25519	497.6	2,010

Provenance

file	rows	sha256
primitives.csv	216	52ef3b82ef1a08ce...
handshake.csv	60	4e253ac5d0cc7737...
environment.csv	13	9d8e3abace057568...

Notes

- QudoSSL runs in **FIPS mode**; the FIPS tables are the primary figures. Under the FIPS provider **key generation is materially slower** (mandatory pairwise-consistency test); encapsulation, decapsulation, signing and verification match the default-provider rates.
- Wall-clock and **indicative**: CPU frequency, core type and scheduling all affect it. The handshake harness runs in memory with no network, so its rate is an upper bound, not a deployment throughput.